# Trust-Aware Weight Calculation for the Distributed Observer

## 1 Purpose

This note summarizes the weighting rule used by the trust-aware distributed observer. The presentation is intentionally simpler than the software implementation: it keeps the main mathematical structure and collects secondary implementation safeguards into compact terms. The objective is to obtain a row-stochastic set of weights for each host vehicle $i$ and target vehicle $j$, so that reliable direct measurements and trusted neighbor estimates receive higher influence while unreliable information is automatically attenuated.

## 2 Notation

Let $i$ denote the host vehicle that runs the observer and let $j$ denote the target vehicle being estimated. At time $k$, $\hat{\boldsymbol{x}}_i^j(k) \in \mathbb{R}^n$ is the current estimate of target $j$ stored by host $i$. The target's direct V2V measurement is denoted by $\boldsymbol{z}_j(k)$, and $\hat{\boldsymbol{x}}_\ell^j(k)$ denotes neighbor $\ell$'s estimate of target $j$.

The trust model provides two types of scores:

- $T_{i,\ell}(k) \in [0,1]$: host $i$'s trust in neighbor $\ell$ as an information source.
- $L_{i,j}(k) \in [0,1]$: local/direct trust in target $j$'s broadcast measurement.

The design parameters are the trust threshold $\theta_T$, the maximum number of neighbors $\kappa$, the nominal direct-measurement gain $a_0$, the nominal self/prediction gain $a_s$, the nominal neighbor budget $a_N = 1 - a_0 - a_s$, and the maximum per-neighbor weight $w_{\max}$. In the current configuration, typical values are $a_0 = 0.4$, $a_s = 0.2$, $a_N = 0.4$, $\theta_T = 0.5$, and $w_{\max} = 0.4$.

## 3 Trusted Information Set

For each target $j$, host $i$ first selects the neighbors that both provide an estimate of $j$ and pass the trust threshold:

$$\mathcal{L}_i^j(k) = \mathrm{Top}_\kappa \left\{ \ell \in \mathcal{N}_i \setminus \{j\} \;\middle|\; \hat{\boldsymbol{x}}_\ell^j(k) \text{ is available},\ T_{i,\ell}(k) \geq \theta_T \right\}, \tag{1}$$

where $\mathrm{Top}_\kappa\{\cdot\}$ keeps at most $\kappa$ neighbors, sorted by descending trust. The target $j$ itself is not used as a neighbor source for estimating $j$; its information enters only through the direct measurement channel.

The selected neighbor trust scores are normalized as

$$b_{i,\ell}^j(k) = \frac{T_{i,\ell}(k)}{\sum_{r \in \mathcal{L}_i^j(k)} T_{i,r}(k)}, \qquad \ell \in \mathcal{L}_i^j(k), \tag{2}$$

when the denominator is nonzero. These normalized values determine how the neighbor budget is distributed.

## 4 Raw Channel Scores

The observer uses three information channels: direct measurement, self prediction, and neighbor consensus. Define

$$\chi_{i,j}(k) = \begin{cases} 1, & \text{if the direct measurement } \boldsymbol{z}_j(k) \text{ is available and accepted,} \\ 0, & \text{otherwise.} \end{cases}$$

The raw channel scores are

$$\bar{w}^j_{0,i}(k) = a_0\, \chi_{i,j}(k)\, L_{i,j}(k)\, \phi^j_0(k), \tag{3}$$

$$\bar{w}^j_{s,i}(k) = a_s\, \rho^j_{s,i}(k)\, \phi^j_s(k), \tag{4}$$

$$\bar{w}^j_{i,\ell}(k) = a_N\, b^j_{i,\ell}(k)\, \phi^j_N(k), \qquad \ell \in \mathcal{L}^j_i(k). \tag{5}$$

Here $w_0$ weights the direct measurement, $w_s$ is the residual self/prediction weight, and $w^j_{i,\ell}$ weights neighbor $\ell$'s estimate of target $j$. The factors $\phi^j_0, \phi^j_s, \phi^j_N$ are compact confidence multipliers derived from the attack flags; they are equal to one during normal operation. The self confidence $\rho^j_{s,i} \in [0,1]$ can be chosen from the self-consistency score of the trust model. A practical choice is

$$\rho^j_{s,i}(k) = \gamma_{\min} + (1 - \gamma_{\min})\gamma^j_{s,i}(k), \tag{6}$$

where $\gamma^j_{s,i} \in [0,1]$ is the self-consistency score and $\gamma_{\min}$ prevents the self channel from vanishing.

## 5 Normalization and Influence Cap

The raw scores are normalized jointly:

$$S^j_i(k) = \bar{w}^j_{0,i}(k) + \bar{w}^j_{s,i}(k) + \sum_{\ell \in \mathcal{L}^j_i(k)} \bar{w}^j_{i,\ell}(k). \tag{7}$$

If $S^j_i(k) > 0$, the preliminary weights are

$$\tilde{w}^j_{q,i}(k) = \frac{\bar{w}^j_{q,i}(k)}{S^j_i(k)}, \qquad q \in \{0, s\} \cup \mathcal{L}^j_i(k). \tag{8}$$

If $S^j_i(k) = 0$, the observer falls back to the self channel: $w^j_{s,i}(k) = 1$.

To prevent one neighbor from dominating the update, each neighbor weight is capped:

$$w^j_{i,\ell}(k) \le w_{\max}, \qquad \ell \in \mathcal{L}^j_i(k). \tag{9}$$

Any excess weight is redistributed among the remaining uncapped neighbors in proportion to $b^j_{i,\ell}$. If no uncapped neighbor remains, the excess is assigned to the self channel. Finally,

$$w^j_{s,i}(k) = 1 - w^j_{0,i}(k) - \sum_{\ell \in \mathcal{L}^j_i(k)} w^j_{i,\ell}(k), \tag{10}$$

which gives the row-stochastic property

$$w^j_{0,i}(k) + w^j_{s,i}(k) + \sum_{\ell \in \mathcal{L}^j_i(k)} w^j_{i,\ell}(k) = 1. \tag{11}$$

## 6 Observer Update

Using the weights above, the consensus-corrected estimate is written as

$$\boldsymbol{x}_{c,i}^{j}(k) = w_{s,i}^{j}(k)\hat{\boldsymbol{x}}_{i}^{j}(k) + w_{0,i}^{j}(k)\boldsymbol{z}_{j}(k) + \sum_{\ell \in \mathcal{L}_{i}^{j}(k)} w_{i,\ell}^{j}(k)\hat{\boldsymbol{x}}_{\ell}^{j}(k). \tag{12}$$

Equivalently, this can be written in correction form as

$$\boldsymbol{x}_{c,i}^{j}(k) = \hat{\boldsymbol{x}}_{i}^{j}(k) + w_{0,i}^{j}(k)\left(\boldsymbol{z}_{j}(k) - \hat{\boldsymbol{x}}_{i}^{j}(k)\right) + \sum_{\ell \in \mathcal{L}_{i}^{j}(k)} w_{i,\ell}^{j}(k)\left(\hat{\boldsymbol{x}}_{\ell}^{j}(k) - \hat{\boldsymbol{x}}_{i}^{j}(k)\right). \tag{13}$$

The corrected state is then propagated through the target dynamics:

$$\hat{\boldsymbol{x}}_{i}^{j}(k+1) = f_{j}\left(\boldsymbol{x}_{c,i}^{j}(k), \boldsymbol{u}_{j}(k), T_{s}\right), \tag{14}$$

where $f_j(\cdot)$ is the discrete-time vehicle model and $T_s$ is the sampling period.

## 7 Paper-Style Simplified Variant

For a more compact theoretical presentation, the same idea can be expressed as an equal-weight rule over a legitimate information set. Define

$$\mathcal{T}_{i}^{j}(k) = \mathcal{L}_{i}^{j}(k) \cup \{0 \mid \chi_{i,j}(k) = 1,\ L_{i,j}(k) \geq \theta_{T}\}, \tag{15}$$

where the index 0 denotes the direct measurement channel. Let

$$n_{i}^{j}(k) = \max\left\{\kappa,\ |\mathcal{T}_{i}^{j}(k)| + 1\right\}. \tag{16}$$

Then every legitimate external channel receives

$$w_{q,i}^{j}(k) = \frac{1}{n_{i}^{j}(k)}, \qquad q \in \mathcal{T}_{i}^{j}(k), \tag{17}$$

and the remaining mass is assigned to the self channel:

$$w_{s,i}^{j}(k) = 1 - \sum_{q \in \mathcal{T}_{i}^{j}(k)} w_{q,i}^{j}(k). \tag{18}$$

This form is less adaptive than the trust-proportional rule, but it is useful for stability analysis because all accepted external sources have bounded and uniform influence.

## 8 Summary

The implemented observer can therefore be interpreted as a trust-weighted convex fusion rule. Direct measurements are weighted by their local trust, neighbor estimates are weighted by source trust, every neighbor is capped to avoid dominance, and the remaining mass stays with the host's own predicted state. This gives a simple, row-stochastic observer update that is suitable for both implementation and academic presentation.